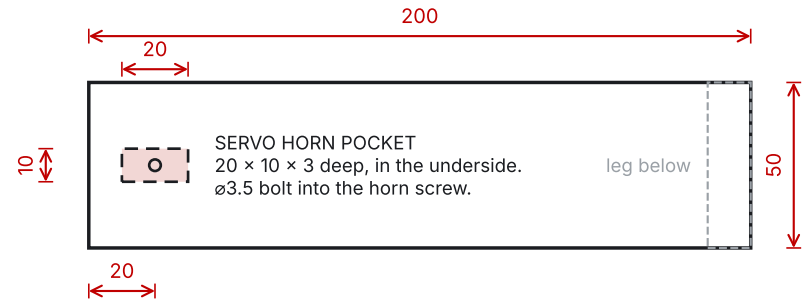
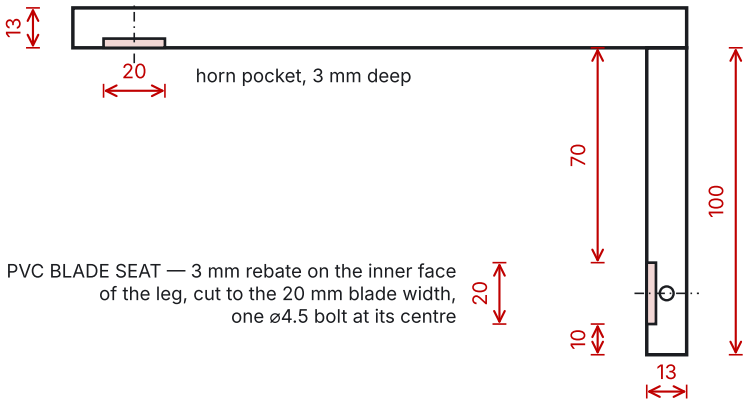
PART A — moving arm: servo horn to PVC deflector blade

All dimensions in millimetres. Material: PLA. Quantity: 4.

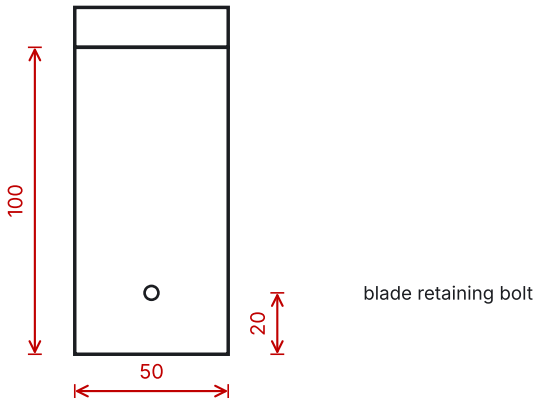
TOP VIEW



SIDE VIEW



FRONT VIEW (leg)

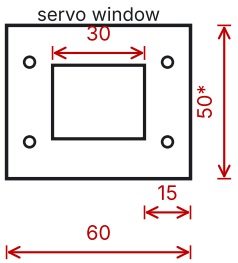


Dimensions from the author's dimensioned sketch of 26 October 2026 and his markup of this drawing the same day, which moved the blade seat to the inner face of the leg and added the pocket that seats the rectangular servo horn. Not on either, and therefore assumed: the servo bolt diameter (ø3.5, M3 clearance) and the blade bolt diameter (ø4.5). The original AutoCAD model was lost; this is a parametric re-derivation and has not been re-printed or re-fitted.

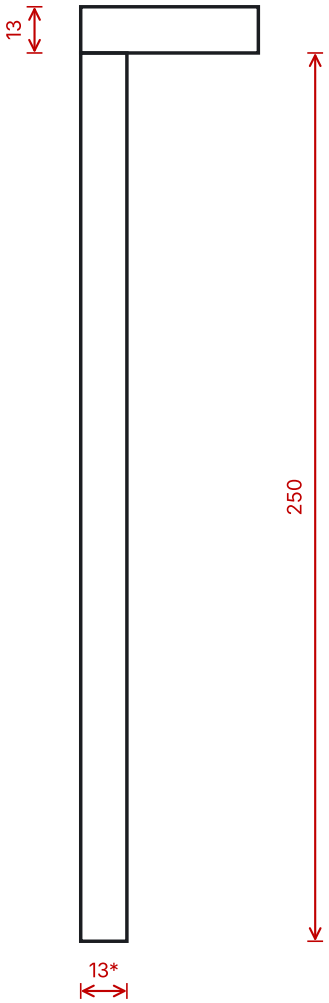
PART B — fixed bracket: servo to conveyor frame

All dimensions in millimetres. Material: PLA. Quantity: 4.

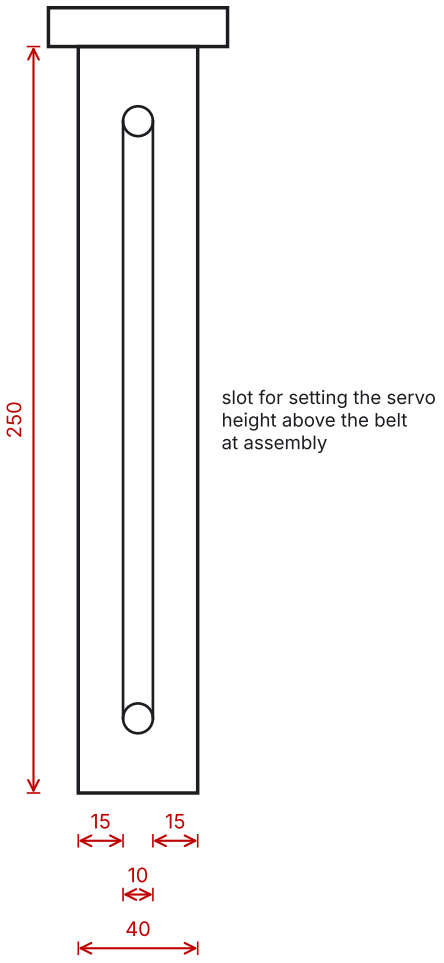
TOP VIEW



SIDE VIEW



FRONT VIEW



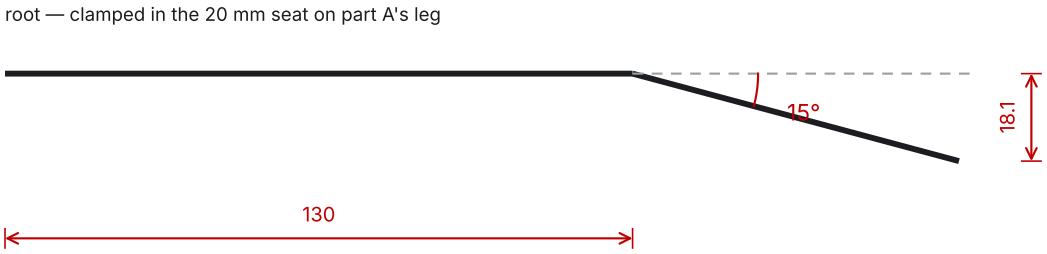
Dimensions marked * are NOT on the author's sketch and are assumed: the plate depth (50 mm) and the post thickness (13 mm).
Also assumed, and not dimensioned here: the servo window depth (24 mm, MG995 body plus clearance), the four bolt holes (ø3.5 at 26 mm pitch) and where the slot stops at each end (20 mm). The original AutoCAD model was lost; this is a parametric re-derivation.

PVC DEFLECTOR BLADE — cut and bend

All dimensions in millimetres. Material: PVC trim profile. Quantity: 4.

FLAT PATTERN — cut from PVC trim profile

SIDE PROFILE — after bending



tip stands 18.1 mm off the straight run

METHOD

1. Cut a 200 × 20 mm strip from PVC trim profile.
2. Drill one ⌀4.5 mm hole on the centre line, 10 mm from the root end.
3. Bend the last 70 mm through 15°, away from the belt. The tip then stands 18.1 mm off the straight run.
4. Bolt the root into the 20 mm seat on part A's leg.

The 200 mm length, the 20 mm width, the bend from 130 mm to the tip and the 15° angle are the author's measurements of the physical blade. Marked * and therefore assumed: the bolt hole position (10 mm from the root). The bend radius is not specified — it was formed by hand, so a builder should match the angle, not a radius.